Mathematical Analysis 2
Seminar 10 – supplementary exercises

Solved exercises: line integrals of the first kind

Problem 1. Compute $\displaystyle\int_C y\cdot e^{-x}\,ds$, where $C:\begin{cases} x=\ln(t^2+1) \\ y=1-t+2\operatorname{arctg} t \end{cases}$, $t\in[0,1]$

Solution.

$$x'(t)=\frac{2t}{t^2+1},\qquad y'(t)=-1+\frac{2}{t^2+1}=\frac{1-t^2}{t^2+1},$$

$$ds=\sqrt{(x'(t))^2+(y'(t))^2}\,dt=\sqrt{\frac{4t^2+(t^4-2t^2+1)}{(t^2+1)^2}}\,dt=dt,\qquad y\cdot e^{-x}=\frac{1-t+2\arctan t}{t^2+1}.$$

It follows that

$$\int_C y\cdot e^{-x}\,ds=\int_0^1\frac{1-t+2\operatorname{arctg} t}{t^2+1}\,dt=\int_0^1\frac{1}{t^2+1}\,dt-\frac{1}{2}\int_0^1\frac{2t}{t^2+1}\,dt+\int_0^1 2\arctan t\cdot(\arctan t)'\,dt$$
$$=\left(\arctan t-\frac{1}{2}\ln(t^2+1)+\arctan^2 t\right)\Big|_0^1=\frac{\pi}{4}-\frac{\ln 2}{2}+\frac{\pi^2}{16}.$$

Problem 2. Compute $\displaystyle\int_C (x+y+z)\,ds$, where C is the line segment $[AB]$, with $A(-1,2,1)$ and $B(2,3,1)$.

Solution. The segment $[AB]$ may be parametrized as follows:

$$\begin{cases} x(t)= & x_A\cdot(1-t)+x_B\cdot t &=(-1)(1-t)+2t &=3t-1 \\ y(t)= & y_A\cdot(1-t)+y_B\cdot t &=2(1-t)+3t &=t+2 \\ z(t)= & z_A\cdot(1-t)+z_B\cdot t &=(1-t)+t &=1 \end{cases},\qquad t\in[0,1]$$

$$ds=\sqrt{(x'(t))^2+(y'(t))^2+(z'(t))^2}\,dt=\sqrt{3^2+1^1+0^2}\,dt=\sqrt{10}\,dt$$

Since $x+y+z=(3t-1)+(t+2)+1=4t+2$, it follows that

$$\int_C(x+y+z)\,ds=\int_0^1(4t+2)\sqrt{10}\,dt=\sqrt{10}\left(2t^2+2t\right)\Big|_0^1=4\sqrt{10}.$$

Problem 3. Compute the length of the curve $C: y=x^2,\quad x\in[0,1]$.

Solution. A parametrization of C is:

$$\begin{cases} x(t)=t \\ y(t)=t^2 \end{cases},\quad t\in[0,1]\qquad ds=\sqrt{(x'(t))^2+(y'(t))^2}\,dt=\sqrt{1+4t^2}\,dt$$

hence the length of C is:

$$l(C)=\int_C ds=\int_0^1\sqrt{1+4t^2}\,dt=\frac{1}{4}\left[\ln\left(2t+\sqrt{4t^2+1}\right)+2t\sqrt{4t^2+1}\right]\Big|_0^1=\frac{1}{4}\left[\ln\left(\sqrt{5}+2\right)+2\sqrt{5}\right].$$

Problem 4. Compute the length of the circle or radius R.

Solution. Using the polar parametrization:

$$\begin{cases} x(t) = R\cos t \\ y(t) = R\sin t \end{cases}, \quad t \in [0, 2\pi]\,; \qquad \mathrm{d}s = \sqrt{(x'(t))^2 + (y'(t))^2}\,\mathrm{d}t = R\,\mathrm{d}t$$

it follows that the length of the circle is $\displaystyle\int_C \mathrm{d}s = \int_0^{2\pi} R\,\mathrm{d}t = 2\pi R.$

Problem 5. Compute the mass of the curve $C : x = \cosh t,\ y = \sinh t,\ z = t,\ t \in [0, \ln 2]$ with the density function $\rho(x, y, z) = x$.

Solution. Recall that

$$\cosh t = \frac{\mathrm{e}^t + \mathrm{e}^{-t}}{2}, \quad \sinh t = \frac{\mathrm{e}^t - \mathrm{e}^{-t}}{2}, \quad \cosh^2 t - \sinh^2 t = 1.$$

Since

$$x' = \sinh t, \quad y' = \cosh t, \quad z' = 1$$

$$\mathrm{d}s = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}\,\mathrm{d}t = \sqrt{\cosh^2 t + \sinh^2 t + 1}\,\mathrm{d}t = \sqrt{2\cosh^2 t}\,\mathrm{d}t = \sqrt{2}\cosh t\,\mathrm{d}t,$$

it follows that

$$\int_C \rho(x, y, z)\,\mathrm{d}s = \sqrt{2}\int_0^{\ln 2} \cosh^2 t\,\mathrm{d}t = \frac{\sqrt{2}}{4}\int_0^{\ln 2} \left(\mathrm{e}^{2t} + \mathrm{e}^{-2t} + 2\right)\mathrm{d}t = \ldots = \frac{\sqrt{2}}{32}\,(8\ln 2 + 15)\,.$$